Mini Project Report

ENERGY TRADING DAPP

Project Title: Energy Trading Decentralized Application (DApp)
Subject Code: [Your Subject Code]
Subject Name: Blockchain Technology

Team Members:

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Department: Computer Science Engineering
Year: 3rd Year

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1. Introduction

Energy trading is a crucial component in modern smart grid systems. Traditional centralized systems suffer from inefficiencies, lack of transparency, and dependency on intermediaries.

This project proposes a decentralized energy trading system using blockchain technology, enabling peer-to-peer energy exchange between users without intermediaries.

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2. Problem Statement

Traditional energy trading systems face:

- Lack of transparency
- Centralized control
- High operational costs
- Inefficient peer-to-peer exchange

Solution: Build a decentralized application using blockchain to enable transparent and secure energy trading.

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3. Objectives

- Develop a decentralized energy trading platform

- Enable peer-to-peer transactions
- Ensure transparency and immutability
- Integrate blockchain with a user-friendly UI
- Provide real-time balance and energy tracking

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4. Functional Requirements

- User wallet connection (MetaMask)
- Add balance (ETH)
- Add energy units (seller)
- Buy energy from another user
- Display real-time balance and energy
- Show transaction confirmations

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5. Type of Blockchain

Private Blockchain (Local Hardhat Network)

Justification:

- Controlled environment for development
- Faster transactions
- No real gas cost
- Suitable for academic projects

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6. Consensus Algorithm

Proof of Authority (PoA) – Simulated via Hardhat

Justification:

- Faster block validation
- No mining required
- Suitable for local testing
- Efficient for small-scale systems

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7. Technology Stack

- **Blockchain:** Ethereum

- **Smart Contracts:** Solidity
- **Development Framework:** Hardhat
- **Frontend:** React.js
- **Library:** Ethers.js
- **Wallet:** MetaMask
- **Language:** JavaScript

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8. Smart Contract Design

Contract: EnergyTrading.sol

Functions:

1. addBalance()

- Payable function
- Adds ETH to user balance

2. addEnergy(uint units)

- Adds energy units for seller

3. buyEnergy(address producer, uint units)

- Transfers energy from seller to buyer
- Transfers balance accordingly

4. getBalance(address user)

- Returns user balance

5. getEnergy(address user)

- Returns energy units

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9. Implementation Details

Network Setup

- Hardhat local blockchain started using:

```
npx hardhat node
```

Smart Contract Deployment

```
npx hardhat run scripts/deploy.js --network localhost
```

API Integration

- Used **Ethers.js** to interact with smart contract

- Contract instance connected using MetaMask signer

UI Implementation

- React-based interface
- Buttons for:
    - Add Balance
    - Add Energy
    - Buy Energy
- Real-time data fetched using contract calls

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10. Testing and Validation

Test cases performed:

- Wallet connection successful
- Balance addition verified
- Energy addition verified
- Energy purchase validation
- Error handling (insufficient balance/energy)

All test cases executed successfully.

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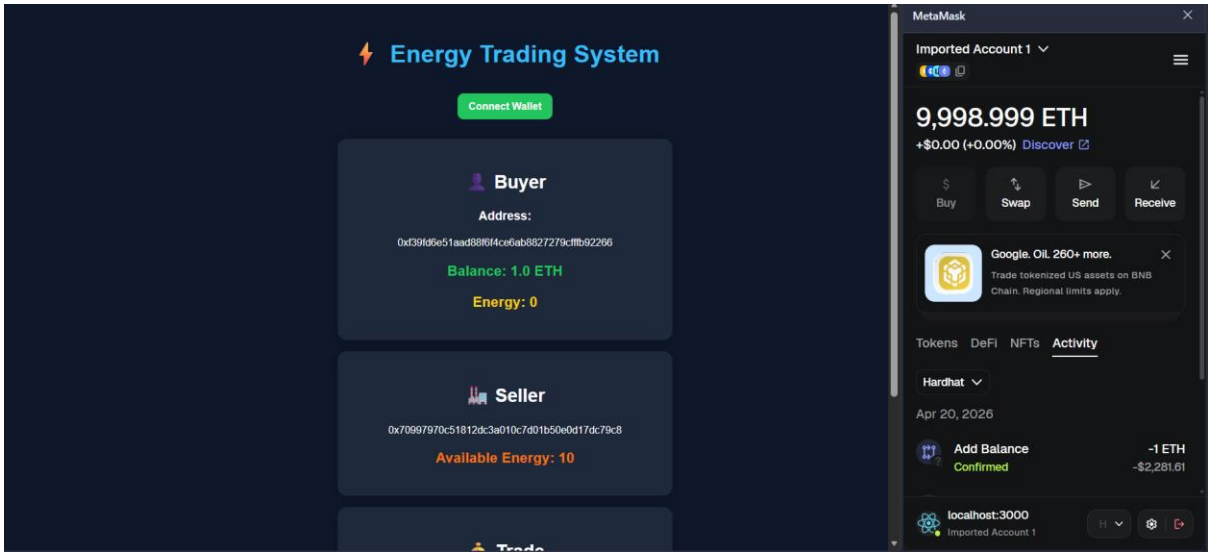
11. Results and Discussion

Observations:

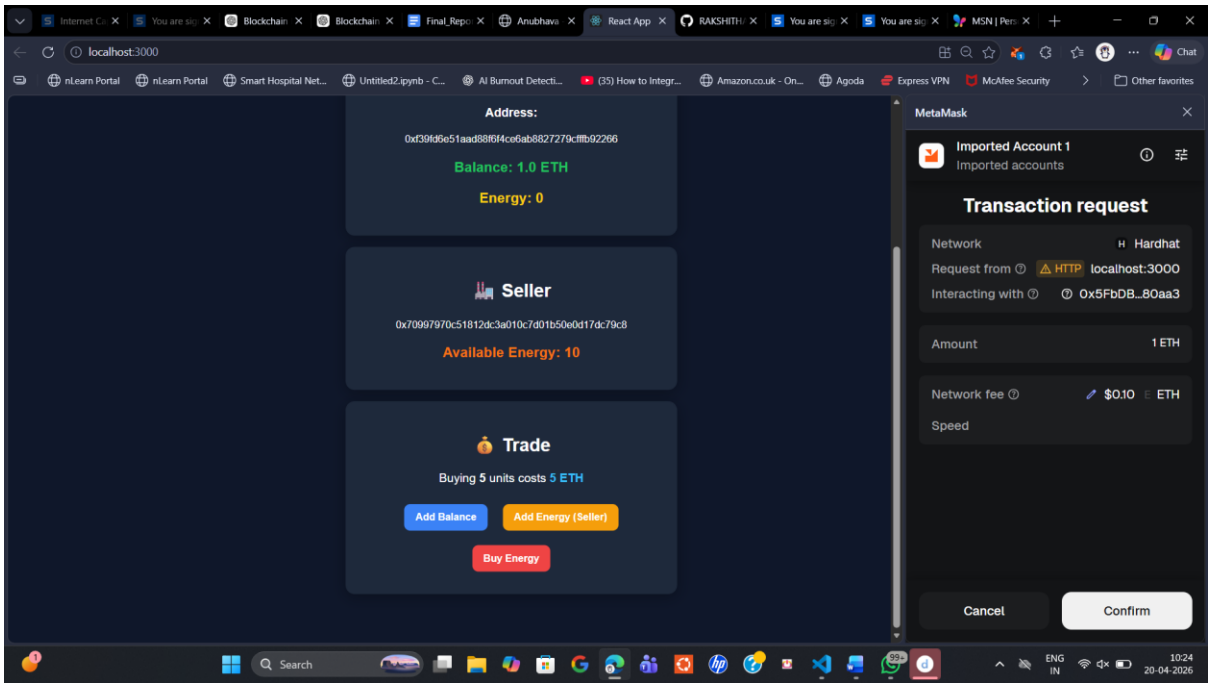
- Smart contract executed correctly
- Real-time UI updates working
- Transactions confirmed via MetaMask
- Buyer and seller balances updated correctly

Screenshots to include:

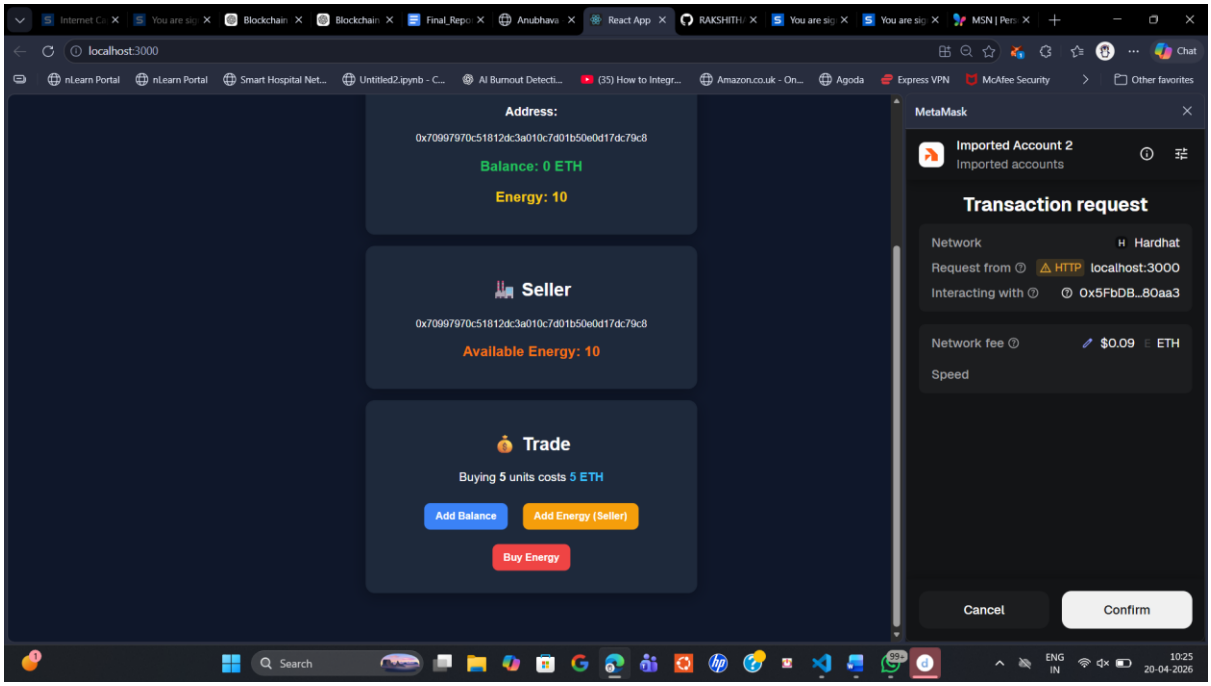
- Wallet connection



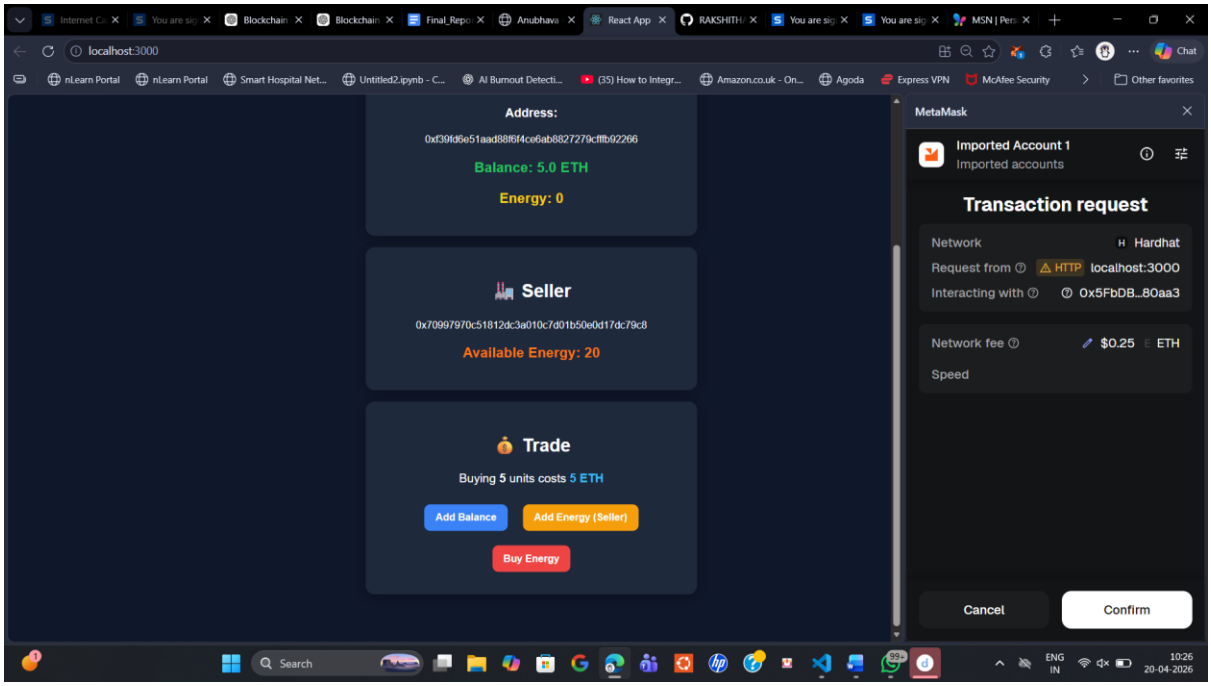
- Add balance transaction



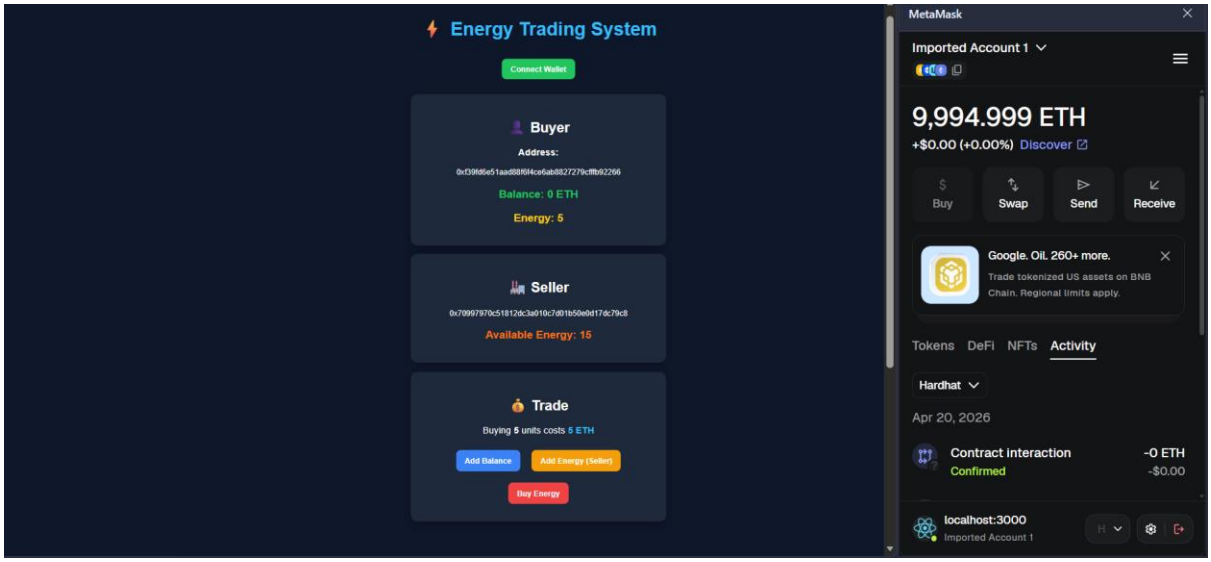
- Add energy

- 

- Buy energy



- Final UI state



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12. Security Analysis

- **Smart contract validation using require()**
- Prevents:
  - Buying without sufficient balance
  - Selling without sufficient energy
- Blockchain ensures:
  - Immutability
  - Transparency
  - Tamper-proof transactions

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13. GitHub Repository

https://github.com/RAKSHITHA9990/energy-trading-dapp

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14. Conclusion

This project successfully demonstrates a decentralized energy trading system using blockchain technology. It eliminates intermediaries, improves transparency, and enables secure peer-to-peer transactions.

Future improvements can include:

- Deployment on testnet (Sepolia)
- Advanced UI
- Real-time energy pricing

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15. References

- Ethereum Documentation
- Solidity Docs
- Hardhat Documentation
- Ethers.js Documentation
- MetaMask Docs